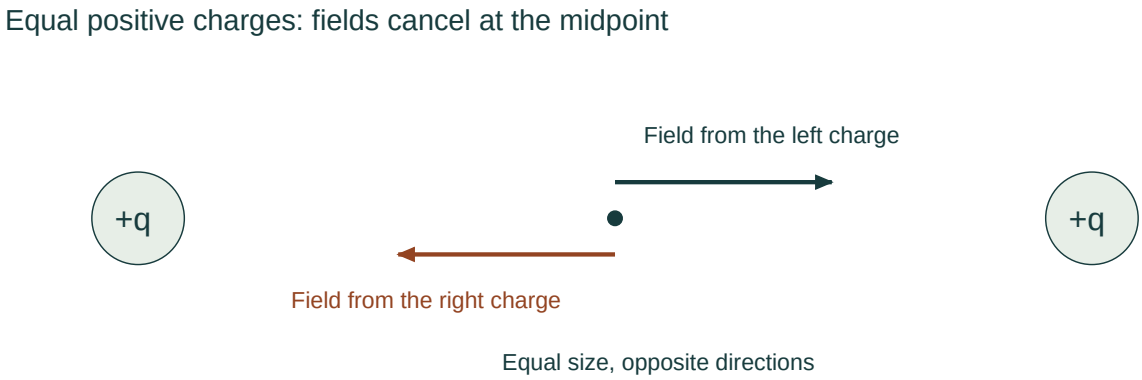
LEVEL 3 / ENGLISH STUDY EDITION

# Electricity and Magnetism

Concept-first coverage of the course plan. Chapters 23-31 have lecture support; inductance, AC and EM waves are course-plan extensions without separate lecture files in this folder. Advanced field integrals are recognition topics, not worked derivations.




### A clear route from idea to answer

Concept explanations, symbols and conditions, worked steps, recall checks, and two practice tests with explained solutions.

12 lessons | Revised 1 November 2026 | Beginner to exam practice

### How to use this handout

1. Read the concept and check the symbol meanings before choosing a formula.
2. Cover the worked steps and reproduce the reasoning yourself.
3. Attempt the quick check and the two subject tests without notes.
4. Mark your work; record whether the error was concept, method, algebra, units or reading. Retry after a delay.

### Your study route

1. Charges, conductors and Coulomb force
2. Electric field and superposition
3. Flux and Gauss law
4. Symmetry and conductors
5. Potential is not field
6. Capacitors and dielectrics
7. Current, resistance and electrical power
8. DC networks, Kirchhoff rules and RC behaviour
9. Magnetic force: direction before calculation
10. Sources of magnetic fields
11. Induction and Lenz law
12. Inductance, AC and electromagnetic waves

---

**LESSON 01 / 12**

## Charges, conductors and Coulomb force

### After this lesson, you can

- Predict attraction or repulsion from charge signs.
- Use simple Coulomb-force ratios without unnecessary arithmetic.

Start with: Positive and negative charge; squares; micro means one millionth.

**THEORY**

Charge is conserved and occurs in integer multiples of the elementary charge. Ordinary charging transfers electrons. Like charges repel and unlike charges attract. Conductors let charge move readily; insulators restrict it. A neutral object can still be attracted by polarization.

## RULE

$$F = k|q_1 q_2|/r^2$$

$$k = 8.99 \times 10^9 \text{ N m}^2/\text{C}^2$$

$F$ electrostatic force magnitude (N)

$q_1, q_2$ source charges (C)

$r$ separation of charges (m)

### When this applies

Coulomb calculations here use point charges, or a stated model equivalent to point charges.

### A method that earns marks

1. Mark the signs and positions of both charges.
2. Predict attraction or repulsion before calculating magnitude.
3. For a ratio, compare the charge product and the square of the separation.
4. For a numerical force, convert charges and lengths to SI units first.

## WORKED EXAMPLE

Two charges at distance r exert force F. Doubling their separation gives $\frac{F}{4}$. Doubling one charge doubles the force.

1. Keep both charges fixed and replace the separation r by 2r.
2. The denominator becomes $(2\text{r})^2 = 4\text{r}^2$, so the force becomes F/4.
3. If instead one charge doubles at the original separation, its product with the other charge doubles, so force becomes 2F.
4. Changing separation or charge magnitude alone does not change attraction into repulsion.

**Common mistake**
A negative charge does not mean a negative force magnitude.

Exam habit: A magnitude answer is positive; state attraction or repulsion separately to communicate direction.

## SELF-CHECK

Tripling separation with fixed charges changes the force by what factor?

Answer: It becomes one ninth as large.

LESSON 02 / 12

# Electric field and superposition

## After this lesson, you can

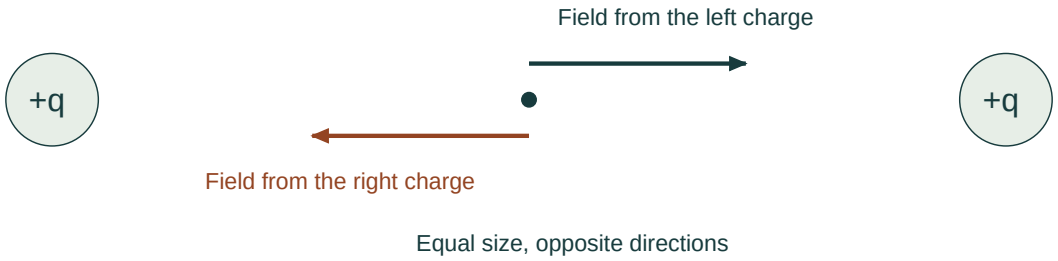
- Combine electric fields by their directions at an observation point.
- Explain why a negative test charge feels force opposite the field.

Start with: Vector addition; positive and negative charges; multiplication.

## THEORY

The electric field gives force per unit positive test charge. Field arrows point away from positive sources and toward negative sources. The field exists whether or not a test charge is present. Add fields as vectors, so equal magnitudes can cancel.

Equal positive charges: fields cancel at the midpoint




At this midpoint, the net field and force on any test charge are zero. Where the net field is nonzero, a negative test charge feels force opposite to it.

## RULE

$$E = F/q_{\text{test}}$$
$$E = k|Q|/r^2$$
$$\vec{F} = q\vec{E}$$

$E$ electric field magnitude (N/C)

$q$ charge experiencing the force (C)

$F$ force magnitude (N)

## A method that earns marks

1. Choose the observation point and draw the field arrow from each source there.
2. Point away from a positive source and toward a negative source.

3. Add signed collinear fields or use vector components when needed.

4. Only after finding the net field, use the test-charge sign to obtain force direction.

### WORKED EXAMPLE

At the midpoint between equal positive charges, the two electric fields oppose and cancel. Between equal opposite charges they reinforce toward the negative charge.

1. At the midpoint between equal positive charges, both source distances are the same, so field magnitudes match.

2. The left positive charge points the field right; the right positive charge points it left. The net field is zero.

3. Replace the right source by an equal negative charge: its field now points toward it, also to the right.

4. The two fields therefore reinforce, pointing from the positive source toward the negative source.

**Common mistake**
Zero net field at a point does not mean each source field is zero.

Exam habit: Draw field arrows before multiplying by a test charge; otherwise a negative test charge can lead to two direction reversals.

### SELF-CHECK

If the net field is zero at a point, is a charge placed there forced in either direction by that field?

Answer: No. The electric force qE is zero there in the stated source-field model.

---

LESSON 03 / 12

## Flux and Gauss law

### After this lesson, you can

- Use the surface normal to determine electric flux.
- Distinguish enclosed charge from outside charge in Gauss law.

Start with: Surface area; normal means perpendicular; cosine at 0ř and 90ř.

### THEORY

Electric flux measures the field passing through an oriented surface. The angle is measured from the surface normal, not the surface itself. Net flux through a closed surface depends on enclosed charge. An outside charge can affect the field at the surface but contributes zero net closed-surface flux.

### RULE

$$\Phi = EA\cos\theta$$

$$\Phi_{\text{closed}} = Q_{\text{enclosed}}/\epsilon_0$$

$\Phi$ electric flux (N $m^2$/C)

$E$ electric field magnitude (N/C)

$A$ surface area ($m^2$)

$Q_{\text{enclosed}}$ net charge inside a closed surface (C)

### A method that earns marks

1. Decide whether the question concerns one flat surface or a complete closed surface.
2. For a flat surface, draw the normal and identify its angle to the field.
3. For a closed surface, add only the enclosed charges with signs.
4. Interpret positive closed flux as net outward flux and negative as net inward flux.

**WORKED EXAMPLE**

A cube enclosing +q has net outward flux $\frac{q}{\varepsilon_0}$. Moving q off-centre changes local field values but not the total flux.

1. The cube is a closed surface enclosing net charge +q.
2. Gauss law fixes its total outward flux as q divided by $\epsilon$0.
3. Moving q off-centre leaves enclosed charge unchanged, so total flux is unchanged.
4. The nearby faces can now have stronger local fields; Gauss law does not say each face receives an equal share after the move.

**Common mistake**
Zero net closed-surface flux does not imply zero field everywhere.

Exam habit: The angle in EA cos $\theta$ is to the normal; a field parallel to the surface gives zero flux.

**SELF-CHECK**

Can an external charge produce a nonzero field on a box while contributing zero net flux through it?

Answer: Yes. Field enters and leaves; the total closed-surface flux from that external charge cancels.

---

LESSON 04 / 12

## Symmetry and conductors

### After this lesson, you can

- State the electrostatic properties of conducting material.
- Use the inverse-square field rule outside a spherical conductor.

Start with: Electric field; conductor versus insulator; inverse-square ratios.

## THEORY

Gauss law is always valid, but it easily gives field magnitude only when symmetry makes the surface calculation simple. In electrostatic equilibrium the field inside conducting material is zero and excess charge lies on its surface. A conductor is an equipotential. For the outside-sphere inverse-square rule, assume an isolated conducting sphere with no external fields.

## RULE

$$E_{\text{outside sphere}} = k|Q|/r^2$$

$$(r > R)$$

$Q$ net sphere charge (C)

$R$ sphere radius (m)

$r$ distance from sphere centre (m)

$E$ electric field magnitude (N/C)

### When this applies

The sphere is isolated with spherical symmetry and is in electrostatic equilibrium.

### A method that earns marks

1. Check that the conductor is in electrostatic equilibrium.
2. Distinguish a point in the conducting material from a point outside the sphere.
3. For an outside point measure distance from the centre, not from the surface.
4. Use the point-charge field or a simple ratio, then describe the radial direction.

## WORKED EXAMPLE

A charged conducting sphere behaves outside like a point charge at its centre. At twice the outside radius the field is one quarter as large.

1. Outside a spherically symmetric charged conducting sphere, the field has the same form as a charge at its centre.
2. Compare two outside distances r and 2r.
3. The square of the distance grows by a factor of four, so the field magnitude becomes E/4.
4. This comparison uses centre distances; doubling the gap from the surface would generally give a different ratio.

**Common mistake**
Do not extend the zero-field-in-metal statement to every cavity containing a charge.

Exam habit: Write r > R beside the outside-sphere formula to show that its domain has been checked.

**SELF-CHECK**

A field is 80 N/C at 0.20 m from the centre. What is it at 0.40 m if both points are outside?

Answer: 20 N/C: the centre distance doubles and the field is divided by four.

---

**LESSON 05 / 12**

# Potential is not field

## After this lesson, you can

- Add scalar potentials with their charge signs.
- Relate potential difference to energy change and field work.

Start with: Scalar versus vector; signed energy changes; potential energy.

**THEORY**

Electric potential is energy per unit charge, a scalar. Electric field is a vector. Add potentials with charge signs, but add fields with directions. Work by the electric field equals the decrease in electric potential energy.

**RULE**

$$V = kQ/r \text{ (zero at infinity)}$$

$$\Delta U = q\Delta V$$

$$W_{\text{field}} = -\Delta U$$

$V$ electric potential (V = J/C)

$\Delta V$ final minus initial potential (V)

$\Delta U$ change in potential energy (J)

$q$ moving charge (C)

## A method that earns marks

1. Separate a field question, requiring direction, from a potential question, requiring signed scalar addition.
2. Choose the potential reference supplied by the problem.
3. Calculate energy change from charge times final-minus-initial potential.
4. Reverse the sign of that energy change to obtain work by the electric field.

**WORKED EXAMPLE**

Equal positive charges can give zero field at their midpoint but positive potential there. Cancellation rules differ for scalars and vectors.

1. At the midpoint of two equal positive charges, equal field arrows point in opposite directions and cancel.
2. Each positive charge contributes a positive potential at that point when zero potential is chosen at infinity.
3. The two potentials add rather than cancel because potential has no directional arrow.
4. Thus zero net field and positive potential can occur at the same point without contradiction.

**Common mistake**
A charge moving to lower potential does not always lose potential energy; a negative charge does the opposite.

Exam habit: Write ΔV = Vfinal - Vinitial before calculating; swapping the endpoints reverses both energy and work signs.

**SELF-CHECK**

A +2 C charge moves through ΔV = +3 V. What are ΔU and work by the field?

Answer: ΔU = +6 J and work by the field = -6 J.

---

**LESSON 06 / 12**

# Capacitors and dielectrics

## After this lesson, you can

- Distinguish series and parallel capacitor connections.
- Calculate stored charge and identify the effect of a dielectric.

Start with: Potential difference; charge; reciprocals and addition.

**THEORY**

A capacitor stores separated charge and electric-field energy. Capacitance describes charge stored per volt. Parallel capacitors share voltage and their capacitances add. Series capacitors carry equal charge magnitude and have a smaller equivalent capacitance.

**RULE**

$$Q = CV$$
$$U = \tfrac{1}{2}CV^2$$
$$C_{\parallel} = C_1 + C_2$$
$$1/C_s = 1/C_1 + 1/C_2$$

$C$ capacitance (F)

$Q$ stored charge magnitude on either plate (C)

$V$ potential difference between plates (V)

$U$ stored electric energy (J)

### A method that earns marks

1. Identify shared nodes: parallel capacitors share voltage, while ideal series capacitors carry equal charge magnitudes.
2. Add capacitances for parallel or add their reciprocals for series.
3. Use Q = CV with compatible prefixes such as ţF times V giving ţC.
4. For dielectric questions first check whether voltage or charge is held fixed.

**WORKED EXAMPLE**

Two 6 $\mu$F capacitors give 12 $\mu$F in parallel and 3 $\mu$F in series. Inserting a dielectric increases capacitance.

1. For two 6 ţF capacitors in parallel, capacitances add: 6 + 6 = 12 ţF.
2. For series, 1/Cs = 1/6 + 1/6 = 1/3 in reciprocal ţF, so Cs = 3 ţF.
3. The series result is smaller than either individual capacitor, providing a quick plausibility check.
4. Inserting a suitable dielectric increases capacitance; the resulting charge or voltage change depends on the connection to a battery.

**Common mistake**
Do not use resistor series/parallel rules for capacitors.

Exam habit: For series, invert the reciprocal sum at the end; stopping at 1/C is a common lost result mark.

**SELF-CHECK**

What charge magnitude is stored by 4 ţF at 12 V?

Answer: 48 ţC, from Q = CV.

### What changes a capacitor's capacitance?

These simple comparisons cover the geometry and dielectric ideas in the lecture without a derivation.

- Use the first formula for parallel plates with a small gap compared with plate size, so edge effects can be ignored.
- Here $A$ is overlapping plate area, $d$ is separation, and $\varepsilon$ describes the material filling the gap.
- Doubling area doubles capacitance. Doubling separation halves it.
- A dielectric filling the gap increases capacitance. Battery attached: voltage stays fixed. Disconnected and isolated: charge stays fixed.

$$C = \frac{\varepsilon A}{d}$$
$$Q = CV$$

An isolated charged capacitor has capacitance 2 $\mu$F and voltage 12 V. A dielectric triples its capacitance. Find the new charge and voltage.

1. Initial charge: $Q = CV = 2 \times 12 = 24\,\mu\text{C}$.
2. Isolation keeps the charge at $24\,\mu\text{C}$.
3. New capacitance: $6\,\mu\text{F}$; new voltage: $V = 24/6 = 4\,\text{V}$.

Charge stays $24\,\mu\text{C}$; voltage falls to $4\,\text{V}$.

---

**LESSON 07 / 12**

# Current, resistance and electrical power

## After this lesson, you can

- Find current, voltage and power for an ohmic resistor.
- Predict how wire length and area change resistance.

Start with: Charge flow; multiplication and division; circuit voltage.

### THEORY

Current is the rate of charge flow. Conventional current follows positive-charge flow, opposite electron drift in a metal. Resistance depends on material and geometry. Ohm law describes an Ωic element under fixed physical conditions.

### RULE

$$I = \Delta Q/\Delta t$$
$$V = IR$$
$$R = \rho L/A$$
$$P = VI$$

$I$ conventional current (A)

$R$ resistance (Ω)

$V$ potential difference (V)

$P$ electrical power (W)

$\rho$ resistivity (Ω m)

## When this applies

Ohm law is used for an ohmic resistor under unchanged physical conditions.

## A method that earns marks

1. Label voltage across the resistor and current through it.
2. Use V = IR to find the missing electrical quantity.
3. Calculate power using VI or an equivalent resistor relation.

4. For wire comparisons hold material and temperature fixed before changing length or area.

### WORKED EXAMPLE

A 6 Ω resistor across 12 V carries 2 A and dissipates 24 W. A longer wire of the same material and area has greater resistance.

1. The 6 Ω resistor has 12 V across it, so I = 12/6 = 2 A.
2. Its electrical power is VI = 12 times 2 = 24 W.
3. Power of 24 W means 24 joules of energy are transferred each second.
4. A longer wire with unchanged material, area and temperature has larger resistance because R is proportional to length.

**Common mistake**
More resistance lowers power at fixed voltage but raises power at fixed current.

Exam habit: Do not confuse power in watts with energy in joules; multiply power by time when energy is requested.

### SELF-CHECK

What happens to resistance if a wire length doubles but material, area and temperature stay fixed?

Answer: Resistance doubles.

---

LESSON 08 / 12

## DC networks, Kirchhoff rules and RC behaviour

### After this lesson, you can

- Reduce simple resistor networks and recover branch currents.
- Describe the initial and final behaviour of an RC charging circuit.

Start with: Ohm law; series and parallel connections; charge and energy conservation.

### THEORY

Series resistors share current; parallel resistors share voltage. Kirchhoff junction rules express charge conservation, and loop rules express energy conservation. An initially uncharged capacitor charges gradually through a resistor and eventually blocks steady DC in the ideal model.

### RULE

$$R_s = R_1 + R_2$$

$$1/R_p = 1/R_1 + 1/R_2$$

$$\tau = RC$$

$R_{eq}$ equivalent resistance (Ω)

$I$ current (A)

$V$ voltage (V)

$\tau$ RC time constant (s)

### A method that earns marks

1. Identify series elements by shared current and parallel elements by shared voltage.
2. Reduce the network and calculate total source current.
3. Return to the original circuit to calculate branch currents or voltage drops.
4. Check junction current sums and loop voltage sums; for RC questions distinguish transient from long-time behaviour.

**WORKED EXAMPLE**

Two 6 Ω resistors in parallel give 3 Ω. Across 12 V the total current is 4 A, split equally into 2 A in each branch.

1. Both 6 Ω resistors are connected across the same two nodes, so they are parallel.
2. Their equal-resistor equivalent is $6/2 = 3$ Ω.
3. The 12 V source therefore supplies $12/3 = 4$ A total.
4. Each branch still has 12 V and carries $12/6 = 2$ A; the branch currents sum to the 4 A source current.

**Common mistake**
Adjacent components in a drawing are not necessarily in series.

Exam habit: After finding an equivalent resistance, redraw or relabel the original circuit before assigning individual voltage drops.

**SELF-CHECK**

After a long time, does an ideal capacitor in a DC series RC circuit carry a steady current?

Answer: No. Once charged, it blocks steady DC in the ideal model.

### One time constant: the 63% and 37% rules

Recognize how quickly a capacitor charges or discharges without deriving an exponential.

- Use an ideal series resistor-capacitor circuit with constant $R$ and $C$. The time constant $\tau$ is measured in seconds.
- Charging from zero: after one time constant, capacitor voltage and charge reach about 63% of their final values.
- Discharging: after one time constant, about 37% of the starting voltage and charge remain.

- A larger resistance or capacitance makes the change slower.

$\tau = RC$

$1 - e^{-1} \approx 0.632$

$e^{-1} \approx 0.368$

A capacitor charges from zero toward 10 V. Its time constant is 2 s. Estimate its voltage after 2 s.

1. The elapsed time is one time constant.
2. Multiply final voltage by 0.632: $10 \times 0.632 = 6.32$ V.

About 6.3 V. It is not fully charged after one time constant.

### Kirchhoff signs and a real battery

Choosing consistent signs is the main skill. A negative current answer tells you the actual direction.

- At a junction, charge does not keep building up: add entering and leaving currents separately.
- Choose a loop direction. Across a resistor, travelling with the assumed current gives $-IR$; travelling against it gives $+IR$.
- Across a battery, going from its negative to positive terminal gives $+\mathcal{E}$; the reverse gives $-\mathcal{E}$.
- For a battery supplying current, internal resistance $r$ causes the terminal voltage to be less than its emf $\mathcal{E}$.

current in = current out

$\sum \Delta V = 0$

$V_{\text{terminal}} = \mathcal{E} - Ir$ (battery delivering current)

A battery has emf 12 V and internal resistance 1 Ω. It supplies a 5 Ω resistor. Find current and terminal voltage.

1. Use one loop: $12 - I(1) - I(5) = 0$.
2. Total resistance is 6 Ω, so $I = 12/6 = 2$ A.
3. Terminal voltage is $12 - (2)(1) = 10$ V. Check the external resistor: $IR = (2)(5) = 10$ V.

Current 2 A; terminal voltage 10 V.

LESSON 09 / 12

# Magnetic force: direction before calculation

## After this lesson, you can

- Determine when magnetic force vanishes or is largest.
- Explain circular motion without a change in kinetic energy.

Start with: Perpendicular directions; circular acceleration; charge sign.

## THEORY

A magnetic field exerts force on a moving charge when its motion has a perpendicular component. The force is perpendicular to velocity and therefore does no work on a point charge. It can change direction without changing speed.

## RULE

$$F = |q|vB\sin\theta$$

$$F_{\text{wire}} = BIL\sin\theta$$

$q$ particle charge (C)

$v$ particle speed (m/s)

$B$ magnetic field magnitude (T)

$F$ magnetic force magnitude (N)

$\theta$ angle between velocity and field (degrees or radians)

## A method that earns marks

1. Decide whether the object is a moving charge or a current-carrying wire.
2. Identify the angle to the magnetic field and the perpendicular motion or wire length.
3. Use the right-hand rule for positive charge or conventional current; reverse for negative charge.
4. Calculate magnitude and check that the force is perpendicular to motion.

## WORKED EXAMPLE

A positive charge moving perpendicular to a uniform magnetic field follows a circular path if no other force acts. The field supplies the inward force.

1. For the positive charge in the example, velocity is perpendicular to the uniform field, so the magnetic force magnitude is qvB.
2. At each point the force is perpendicular to velocity and turns the particle inward.
3. Because the force does no work, speed and kinetic energy remain constant.

4. With no other forces, the continual turning gives a circular path in the plane perpendicular to the field.

**Common mistake**
Do not apply the right-hand direction unchanged to an electron.

Exam habit: A direction answer needs the charge sign; a right-hand-rule result for a positive charge must be reversed for an electron.

**SELF-CHECK**

Does a charge moving exactly along a magnetic field feel magnetic force?

Answer: No. The angle is 0ř or 180ř, so its sine is zero.

---

**LESSON 10 / 12**

# Sources of magnetic fields

## After this lesson, you can

- Relate a current to the direction of the magnetic field it produces.
- Use simple distance and current ratios for a long wire.

Start with: Current; inverse proportionality; circular field-line diagrams.

**THEORY**

Currents produce magnetic fields. Around a long straight wire the field forms circles. Inside an ideal long solenoid the field is approximately uniform. For two parallel wires, currents in the same direction attract and opposite currents repel.

**RULE**

$$B_{\text{wire}} = \mu_0 I/(2\pi r)$$

$$B_{\text{solenoid}} = \mu_0 n I$$

$B$ magnetic field magnitude (T)

$I$ wire current (A)

$r$ perpendicular distance from wire (m)

$n$ solenoid turns per unit length ($\text{m}^{-}$ź)

## When this applies

The wire is sufficiently long and the observation point sufficiently far from its ends for the long-wire approximation.

### A method that earns marks

1. Identify whether the source is a straight wire, a solenoid or a pair of wires.
2. For a straight wire, use the right-hand grip rule to draw field circles.
3. Apply the matching field relation; for ratios hold unstated factors fixed.
4. For paired wires determine attraction or repulsion from relative current directions.

**WORKED EXAMPLE**

Doubling distance from a long wire halves its magnetic field; doubling current doubles the field.

1. For the long wire, B is proportional to I/r.
2. Moving from r to 2r at the same current gives half the original field.
3. Doubling current at fixed r instead doubles the field.
4. If current and distance both double, the two factors cancel and field magnitude stays the same.

**Common mistake**
The wire field varies as inverse distance, not inverse distance squared.

Exam habit: Use 1/r for the long-wire field; the $1/r^2$ rule belongs to a different source geometry.

**SELF-CHECK**

A wire gives 30 ţT at 2 cm. What field does it give at 6 cm with the same current?

Answer: 10 ţT because distance triples.

---

**LESSON 11 / 12**

## Induction and Lenz law

### After this lesson, you can

- Identify which change produces an induced emf.
- Use Lenz law to choose the induced current direction.

Start with: Magnetic field direction; loop area; clockwise versus counterclockwise viewed from a stated side.

**THEORY**

An induced emf requires a change in magnetic flux, which can arise from changing field, area or orientation. Lenz law sets the direction: the induced effect opposes the change of flux. A static magnet near a stationary loop does not produce a sustained induced current.

## RULE

$$|\mathcal{E}| = N|\Delta\Phi_B|/\Delta t$$

$$\Phi_B = BA\cos\theta$$

$\Phi_B$ magnetic flux through one turn (Wb)

$N$ number of coil turns (dimensionless)

$\mathcal{E}$ induced emf (V)

$\Delta t$ time interval of flux change (s)

### A method that earns marks

1. State the original flux direction and whether its magnitude increases or decreases.
2. Choose an induced field that opposes that change.
3. Use the grip rule to convert the required field to current direction, naming the viewing side.
4. For magnitude, multiply the flux change per turn by the number of turns and divide by elapsed time.

## WORKED EXAMPLE

If outward flux increases, the induced field points inward. The current must therefore circulate clockwise when viewed from that side.

1. In the example, outward flux through the loop is increasing.
2. The induced effect must oppose the increase, so its own field points inward.
3. Viewed from the side toward which the original flux points outward, the inward field requires clockwise current.
4. If the outward flux were decreasing instead, the induced field and current direction would both reverse.

**Common mistake**
Opposes the change is not the same as always opposes the original field.

Exam habit: Say oppose the change in flux, not oppose the flux: a decreasing field produces a reinforcing induced field.

## SELF-CHECK

Does a stationary loop near an unchanging stationary magnet have a sustained induced emf?

Answer: No. Its magnetic flux is not changing.

LESSON 12 / 12

# Inductance, AC and electromagnetic waves

## After this lesson, you can

- Convert sinusoidal peak voltage to RMS and calculate resistive heating.
- Relate electromagnetic frequency to vacuum wavelength.

Start with: Square roots; frequency and wavelength; electrical power.

## THEORY

Course-plan extension: an inductor resists changes of current and stores magnetic energy. AC reverses direction periodically. RMS values describe equivalent heating for resistive loads. In vacuum an EM wave has mutually perpendicular electric field, magnetic field and travel direction.

## RULE

$$U_L = \tfrac{1}{2} L I^2$$

$$V_{\text{rms}} = V_{\text{peak}} / \sqrt{2} \text{ (sinusoidal)}$$

$$c = f\lambda \text{ (vacuum)}$$

$V_{\text{peak}}$ maximum sinusoidal voltage magnitude (V)

$V_{\text{rms}}$ root-mean-square voltage (V)

$L$ inductance (H)

$f$ wave frequency (Hz)

$\lambda$ wavelength (m)

## When this applies

The peak-to-RMS conversion assumes a sinusoid; the wavelength calculation assumes propagation in vacuum.

## A method that earns marks

1. Identify whether the question concerns an inductor, sinusoidal AC or an electromagnetic wave.
2. For sinusoidal AC divide peak voltage by the square root of two.
3. Use RMS values for average power in a pure resistor.
4. For vacuum waves calculate wavelength as c divided by frequency and convert metres to the requested unit.

## WORKED EXAMPLE

A sinusoidal supply with peak voltage 141.4 V has RMS voltage 100 V. Light of frequency $6 \times 10^{14}$ Hz has vacuum wavelength 500 nm.

1. For the stated sinusoidal supply, 141.4 divided by $\sqrt{2}$ is approximately 100 V RMS.
2. This RMS value matches the heating effect of 100 V DC in the same pure resistor.
3. For the separate light example, wavelength is 3.00 Œ $10^{8}$ divided by 6 Œ 10ź$^{4}$ = 5 Œ $10^{-7}$ m.
4. Since one nanometre is $10^{-9}$ m, the wavelength is 500 nm.

**Common mistake**
An ideal transformer does not continuously transform steady DC after switching transients.

Exam habit: Keep this extension conceptual: use RMS for resistor power and c = fλ for waves; do not mix peak and RMS values.

## SELF-CHECK

A 20 V peak sinusoid is applied to 10 Ω. What is the average power?

Answer: 20 W: $V_{rms}^2/R = (20^2/2)/10$.

## Before the tests

These original questions practise the lesson objectives. They are not predictions of an exam paper. Use the separate practice book if you want all papers together.

# Electricity and Magnetism / Test A

30 marks | Suggested time: 30 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

## Part I: Choose one answer

1. A neutral object loses electrons. It becomes:

   A. Negative

   B. Positive

   C. Uncharged forever

   D. Magnetic only

2. A negative test charge in a field directed right feels force:

   A. Left

   B. Zero

   C. Right

   D. Up

3. A charge outside a closed empty box gives net flux through the box of:

   A. $\frac{q}{\varepsilon_0}$

   B. Always positive

   C. Always negative

   D. Zero

4. Inside the material of a conductor in electrostatic equilibrium, the electric field is:

   A. Equal to its surface value

   B. Infinite

   C. Zero

   D. Always outward

5. For equal and opposite point charges, take potential as zero at infinity. At their midpoint the potential is:

   A. Zero

   B. Undefined

   C. Positive

   D. Negative

6. Adding a dielectric to an isolated charged capacitor increases C. Its voltage:

   A. Decreases

   B. Increases

C. Must stay fixed

D. Becomes infinite

### Part II: Show your reasoning

1. Two charges exert 12 N at separation r. What is the force at separation 2r, and at r if one charge is tripled?

2. A field of 200 N/C points east. Find the force on a charge of -0.003 C.

3. A uniform field of 50 N/C passes through a flat area of 0.20 square metres. The chosen surface normal points in the same direction as the field. Find flux. What if the normal is perpendicular to the field?

4. An isolated charged conducting sphere, with no external fields, has field magnitude 80 N/C at r = 0.20 m. Find the field at r = 0.40 m. Both distances are from the centre and outside the sphere.

5. A +2 C charge moves through a potential difference of +3 V. Find its potential-energy change and the work done by the field.

6. A 4 $\mu$F capacitor is connected to 12 V. Find the charge in microcoulombs.

# Electricity and Magnetism / Test B

30 marks | Suggested time: 30 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

## Part I: Choose one answer

1. At fixed voltage, doubling resistance makes power:

   A. Unchanged

   B. Double

   C. Half

   D. Four times

2. After a long time in an ideal DC charging circuit, capacitor current is:

   A. Maximum

   B. Negative forever

   C. Zero

   D. Equal to the battery voltage divided by the resistor resistance

3. A charge moving parallel to a magnetic field experiences magnetic force:

   A. $qvB$

   B. Twice $qvB$

   C. Opposite its velocity

   D. Zero

4. Two parallel wires carrying currents in the same direction:

   A. Repel

   B. Attract

   C. Have no interaction

   D. Must carry zero field

5. Which situation produces an induced emf?

   A. No change of area, field or angle

   B. Changing magnetic flux

   C. A stationary unchanging field only

   D. Constant flux

6. In vacuum, electric and magnetic fields of a plane EM wave are:

   A. Parallel to each other

   B. Longitudinal only

C. Zero

D. Perpendicular to each other and to travel

### Part II: Show your reasoning

1. A 3 Ω resistor carries 2 A. Find voltage and power.

2. A 4 Ω and 8 Ω resistor are in series across 24 V. Find total current and each voltage drop.

3. A straight 0.50 m wire carries 2 A perpendicular to a 0.30 T field. Find force magnitude.

4. The field is 30 microtesla at 2 cm from a long straight wire. Find the field at 6 cm at the same current.

5. A 20-turn coil has flux per turn changing by 0.03 Wb in 0.10 s. Find average emf magnitude.

6. A sinusoidal voltage has peak value 20 V. Find its RMS value, then the average power in a 10 Ω resistor.

# Electricity and Magnetism / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

## Part I: MCQs

1. **B.** Removing negative charge leaves a positive net charge.
2. **A.** The negative charge reverses the field direction in $F = qE$.
3. **D.** No charge is enclosed, so total entering and leaving flux cancels.
4. **C.** Free charges redistribute until the internal field is cancelled.
5. **A.** The signed scalar potentials cancel when distances and charge magnitudes match.
6. **A.** Charge remains fixed and $V = \frac{Q}{C}$.

## Part II: Written solutions

1. At 2r: 12/4 = 3 N. Tripling one charge at r: 36 N. Two marks for each ratio and result.
2. Magnitude = 0.003 times 200 = 0.60 N. Direction is west because the charge is negative. Two marks for magnitude and two for direction with reasoning.
3. Parallel normal: flux = 50 times 0.20 = 10 N m squared/C. Perpendicular normal: cosine 90 degrees = 0, so flux = 0. Two marks per case.
4. Inverse-square scaling gives 80 times $(0.2\frac{0}{0}.40)$ squared = 20 N/C. Credit the outside-sphere condition and inverse-square reasoning.
5. Energy change = 2 times 3 = +6 J. Work by field = -6 J. Two marks each; the signs are essential.
6. Q = CV = 4 times 12 = 48 microcoulombs. In SI this is 0.000048 C. Credit formula, substitution, prefix and result.

# Electricity and Magnetism / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

## Part I: MCQs

1. **C.** $P = \frac{V^2}{R}$, so doubling the denominator halves it.
2. **C.** After a long time, the capacitor voltage equals the battery voltage. The resistor voltage and circuit current are zero. During charging, charge moves in the wires onto and off the plates; it does not cross the ideal dielectric.
3. **D.** The sine of the angle is zero.
4. **B.** Each current experiences the magnetic field of the other wire.
5. **B.** Faraday law depends on the rate of flux change.
6. **D.** A plane electromagnetic wave in vacuum is transverse.

## Part II: Written solutions

1. V = IR = 6 V. P = VI = 12 W (or $I^2R$). Two marks per result.
2. Total resistance = 12 Ω; current = 2 A; drops are 8 V and 16 V. One mark each; the drops must sum to 24 V.
3. F = BIL = 0.30 times 2 times 0.50 = 0.30 N. Four marks for formula, perpendicular condition, substitution and unit.
4. B is inversely proportional to r, so B = 30 times 2/6 = 10 microtesla. Credit model, ratio, arithmetic and unit.
5. Magnitude = 20 times 0.03/0.10 = 6 V. The direction needs the direction of the flux change; it cannot be inferred from these magnitudes alone.
6. RMS = 20/square root of 2 = 14.14 V. Average power = RMS squared/10 = 20 W. Two marks per stage.

## Source and coverage notes

Concept-first coverage of the course plan. Chapters 23-31 have lecture support; inductance, AC and EM waves are course-plan extensions without separate lecture files in this folder. Advanced field integrals are recognition topics, not worked derivations.

Source folder: https://drive.google.com/drive/folders/1-MQIsBh0VU5tceb9COmhxqOKTgIaTZ_g
The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. This revised study aid does not replace lecturer-specific derivations or instructions. All practice tests and suggested marking guidance are original study material.